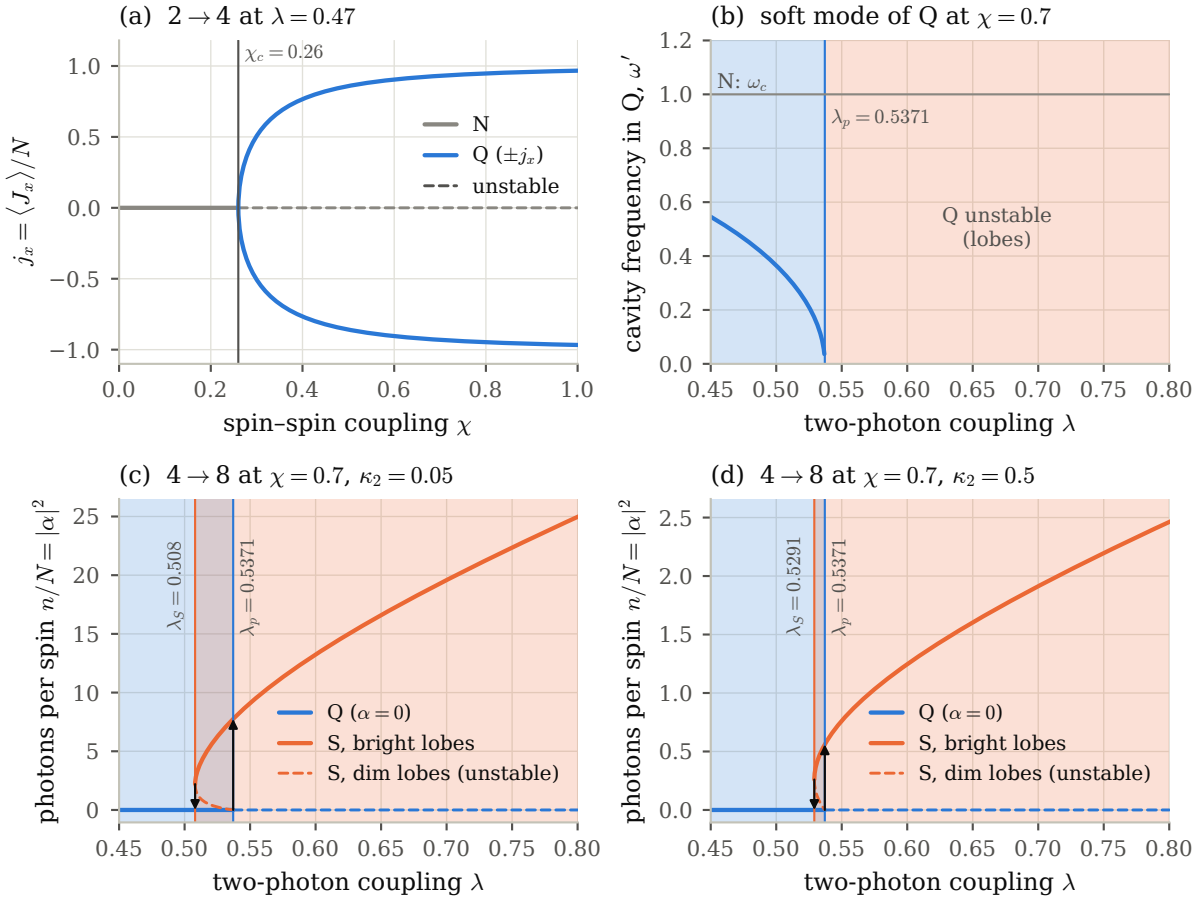
(a) 2 → 4 at λ = 0.47
χc = 0.26
N
Q (±jx)
unstable
jx = ⟨Jx⟩/N
spin–spin coupling χ
(b) soft mode of Q at χ = 0.7
N: ωc
λp = 0.5371
Q unstable
(lobes)
cavity frequency in Q, ω′
two-photon coupling λ
(c) 4 → 8 at χ = 0.7, κ2 = 0.05
λS = 0.508
λp = 0.5371
Q (α = 0)
S, bright lobes
S, dim lobes (unstable)
photons per spin n/N = |α|2
two-photon coupling λ
(d) 4 → 8 at χ = 0.7, κ2 = 0.5
λS = 0.5291
λp = 0.5371
Q (α = 0)
S, bright lobes
S, dim lobes (unstable)
photons per spin n/N = |α|2
two-photon coupling λ